1 $f(x) = x^2 + \frac{16}{x}$

$f'(x) = 2x - \frac{16}{x^2} = 0$

$\frac{2x^3 - 16}{x^2} = 0$

$x^3 = 8$

$\boxed{x = 2}$

$4 + \frac{\cancel{16}^{8}}{\cancel{2}} = 12$

2 $D(x) = (x-1)^2 + (x-4)^2 + (x-9)^2$

!!

[forgot to find value]

$2(x-1) + 2(x-4) + 2(x-9) = 0$

$x = \frac{1+4+9}{3} = \frac{14}{3}$

3

$$L(z) = \ln(1 + e^{z})$$

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

$$L'(z) = \frac{1}{1+e^{z}} \cdot e^{z} = \frac{e^{z}}{1+e^{z}}$$

$$\boxed{L'(z) = \sigma(z)}$$

4

$$y = mx \qquad (1,2) \quad (2,4) \quad (3,5)$$

$$C(m) = (2-m)^2 + (4-2m)^2 + (5-3m)^2$$

$$c'(m) = \cancel{2}(m-2) + \cancel{2}(2m-4) \cdot 2 + \cancel{2}(3m-5) \cdot 3$$

$$6m - 11$$

$$\boxed{m = \frac{11}{6}}$$

5 $f(x) = x^3 - 6x^2 + 9x + 2$

$f'(x) = 3x^2 - 12x + 9$

$f''(x) = 6x - 12$

To find min/max $\rightarrow$

$f'(x) = 0$

$3x^2 - 12x + 9 = 0$

$x^2 - 4x + 3 = 0$

$\boxed{(x-3)(x-1) = 0}$

3 1

To check whether they are min/ma

$f''(1) = 6 - 12 = -6 \rightarrow$ maxima

$f''(3) = 18 - 12 = 6 \rightarrow$ minima

* $f''(x) < 0 \rightarrow$ maxima

$f''(x) > 0 \rightarrow$ minima

6

$y = 1$

$$p = \sigma(z) = \frac{1}{1+e^{-z}}$$

$$L(z) = \text{log-loss} = -\ln(\sigma(z))$$

$$L'(z) = -\frac{1}{\sigma(z)} \cdot \sigma'(z)$$

$$\sigma'(z) = -\frac{1}{(1+e^{-z})^2} \cdot (-1) \cdot e^{-z}$$

$$\boxed{\sigma'(z) = \frac{e^{-z}}{(1+e^{-z})^2}}$$

$$L'(z) = -1 \cdot \cancel{(e^{-z}+1)} \cdot \frac{e^{-z}}{(1+e^{-z})^{\cancel{2}}}$$

$$L'(z) = \frac{-e^{-z}}{1+e^{-z}} = \frac{-\cancel{e^{-z}} \cdot \cancel{e^{z}}}{e^{z}+1} = \frac{-1}{1+e^{z}}$$

$$L'(z) = \frac{-1}{e^{z}+1}$$

$$\sigma(z) - 1 = \frac{1}{1+e^{-z}} - 1$$

$$= \frac{\cancel{e^{z}} - (1) - \cancel{e^{z}}}{1+e^{z}} = -\frac{1}{1+e^{z}}$$

Hence, $\boxed{L'(z) = \sigma(z) - 1}$

7 $f(x) = 2x^2 - 8x + 3$

$f'(x) = 4x - 8 = 0$

$\boxed{x = 2}$

$f(2) = 2.4 - 8.2 + 3$

$= 8 - 16 + 3$

$= -5$

8 $f(x) = x^4 - 4x^3 + 4x^2$

$f'(x) = 4x^3 - 12x^2 + 8x$

$f''(x) = 12x^2 - 24x + 8$

$\Rightarrow f'(x) = 0$

$4x(x^2 - 3x + 2)$

$4x(x-2)(x-1) \to 0$

$\boxed{x = 0, 1, 2}$

$f''(0) = 8$ minima

$f''(1) = -4$ maxima

$f''(2) = 48 - 48 + 8$

$= 8$ minima